

Geometry-Preserving Latent Diffusion in Fixed ChemNet Spaces for Synthetic Ion Mobility Spectra

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Abstract—Latent diffusion in a pre-given latent space can preserve the natural manifold structure of chemical spectral data while matching target distributions. Unlike Variational Autoencoders (VAEs) that enforce Gaussian latent distributions through KL divergence regularization, we argue that pre-known latent spaces occupy complex, non-Gaussian manifolds that reflect the intrinsic geometry of the data. This work introduces a separation-guided latent diffusion approach tailored to decoupled autoencoded embeddings whose target space is predetermined by ChemNet molecular representations. Our approach augments standard noiseprediction objectives with a centroidseparation term and employs a lownoise diffusion schedule designed to retain local manifold geometry while still enabling denoising-based sampling. Using ion mobility spectrometry data across 8 chemicals and 296,692 samples, our approach achieves extraordinary statistical matching (means within 0.08, standard deviations within 7.4%) while preserving complex geometric features that naive Gauss sampling destroys. We further demonstrate that these geometry-preserving synthetic samples provide practical benefits for classifier training.

Index Terms—ion mobility spectrometry, synthetic data, diffusion models, manifold learning, data augmentation

I. Introduction

Generative modeling in scientific domains faces a fundamental challenge: how to create synthetic data that preserves both statistical properties and physically meaningful structure. In ion mobility spectrometry (IMS), data collection is expensive and class imbalance is common, so synthetic generation is attractive. The difficulty is that useful latent representations in this domain are often structured, anisotropic, and class-dependent, rather than simple Gaussian clouds.

Traditional approaches to learned representations, particularly Variational Autoencoders (VAEs), force Gaussian latent distributions through explicit regularization [1]. While this simplifies sampling via $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$, it can overwrite useful geometric information already encoded in the latent space. In contrast, decoupled autoencoders with pre-given embeddings can exhibit elongated tendrils, clusters, and non-convex class geometry that carry chemically relevant information.

A. The Manifold Hypothesis and Latent Representations

Real-world high-dimensional data typically occupies low-dimensional manifolds embedded in latent space [2]. For chemical spectroscopy data, this structure arises

from physical constraints: chemical composition, molecular structure, environmental variation, and measurement physics constrain spectra to a much smaller subspace than the full feature dimension suggests.

Our work builds on a decoupled autoencoder architecture [3] that makes this issue explicit. Rather than learning latent codes from scratch, we fix the latent space to pre-trained ChemNet molecular embeddings [4]. The encoder maps 1676-dimensional IMS spectra into this predetermined 512-dimensional space, and the decoder maps back to spectra. Because ChemNet embeddings were not trained to satisfy a Gaussian prior, the resulting latent geometry is strongly non-Gaussian and often non-convex.

The resulting latent spaces exhibit:

- **Non-Gaussian Geometry:** Elongated structures, tendrils, and clusters reflecting chemical similarity relationships.
- **Chemical Clustering:** Natural separation between distinct chemical classes with smooth transitions within classes.
- **High Inter-Class Variance:** Different chemicals occupy varying extents of latent space, reflected in per-chemical principal component explained variance ratios.

VAE-style Gaussian regularization, while enabling simple sampling, destroys this structure and is incompatible with making use of known latent information. Forcing $q(\mathbf{z}|\mathbf{x}) \approx \mathcal{N}(0, \mathbf{I})$ through KL divergence penalties flattens the manifold, losing the geometric relationships that encode physical meaning.

B. Diffusion in Latent Space

Recent diffusion models provide a practical alternative: instead of constraining the latent distribution, learn to sample from the distribution that is already there [5]. Training diffusion directly in latent space allows generation while preserving manifold geometry.

A key advantage of diffusion models is their ability to work effectively with non-convex manifold structures. As illustrated in Figure 1, while the image manifold itself is non-convex (interpolating between two faces may not yield a valid face), the diffusion process creates a convex space where interpolation is well-defined. By noising two points from the data manifold into the diffusion space,

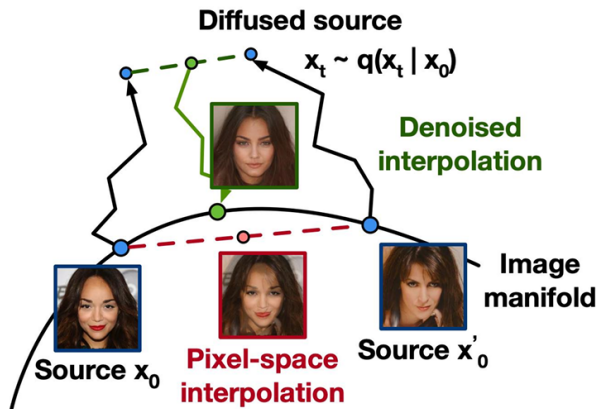


Fig. 1. Manifold structure in diffusion models (adapted from Ho et al. [5]). The image manifold is non-convex: interpolating between two face images may not produce a valid face. However, after adding noise to map points into the diffusion space, the structure becomes convex, allowing meaningful interpolation. The reverse diffusion process then maps interpolated points back to the original manifold, enabling the generation of novel, valid samples. This convexity property is crucial for preserving non-Gaussian manifold structure in latent spaces.

we can perform convex combinations or make a Gaussian draw, then use the reverse diffusion process to map back to valid points on the original manifold. This property makes diffusion models particularly well-suited for generating from complex, non-Gaussian latent distributions.

Our key insight is that geometry-aware training, augmenting standard noise prediction with explicit inter-class geometry regularization, combined with a low-noise schedule, better preserves manifold geometry. This setup maintains distinct chemical clusters while still enabling robust denoising-based generation.

C. Contributions

This work contributes a method for diffusion-based generation in pre-given latent spaces, where the objective is to match an existing distribution without erasing its geometry. The method is designed for decoupled autoencoder settings in which latent coordinates already encode domain knowledge and should be preserved.

We introduce latent diffusion with a joint objective (80% noise prediction, 20% centroid-separation regularization) and class-conditioning on molecular embeddings. This combination keeps inter-class structure from collapsing while still learning a global denoising map that supports generation for multiple chemicals in a single model.

We also show that a low-noise schedule ($\beta_{\max} = 0.02$) is an effective operating point for this problem: it introduces enough stochasticity to support diffusion-based sampling while avoiding aggressive corruption that can blur latent geometry. Empirically, the resulting model preserves non-Gaussian tendrill structure and matches target statistics across 8 chemicals, with additional downstream gains in low-data classifier training.

II. Background

A. Ion Mobility Spectrometry

Ion mobility spectrometry (IMS) is an analytical technique that separates ionized molecules based on their velocity in an electric field [6], [7]. The resulting spectra measure ion intensity as a function of drift time, providing chemical fingerprints for identification and detection applications. IMS offers rapid analysis times (milliseconds) with high sensitivity, making it particularly valuable for chemical detection in security, environmental monitoring, and defense applications [7].

Our IMS dataset comprises 296,692 samples across 8 chemical classes: DEB (Diethylbenzene), DEM (Diethylmalonate), DMMP (Dimethyl methylphosphonate), DPM (Dipropyl methylphosphonate), DtBP (Di-tert-butyl peroxide), JP8 (Jet fuel), MES (Methyl salicylate), and TEPO (Triethyl phosphate). Each sample consists of 1676 features (838 drift time measurements for positive ions, 838 for negative ions), split into 222,519 training/validation and 74,173 test samples.

B. Decoupled Autoencoders with Chemical Embeddings

Following previous work [3], we employ a decoupled autoencoder where the latent space is given ahead of time rather than learned. This is a critical architectural choice that fundamentally shapes the manifold structure we observe.

In standard autoencoders, the latent space is learned end-to-end from reconstruction objectives, with no constraints on its geometry. VAEs impose Gaussian structure through explicit KL regularization. Our decoupled approach takes a third path: we fix the latent space to be the 512-dimensional ChemNet embeddings [4], pre-trained on 3.6 million chemical structures to encode molecular SMILE (Simplified Molecular Input Line Entry System) representations.

The encoder E learns to map IMS spectra $\mathbf{x}_c$ for chemical c to this pre-defined ChemNet embedding space:

$$\mathcal{L}_{\text{enc}} = \|\mathbf{h}_c - E(\mathbf{x}_c)\|^2 \quad (1)$$

where $\mathbf{h}_c \in \mathbb{R}^{512}$ is the fixed ChemNet embedding for chemical c .

The decoder D learns the inverse mapping, reconstructing spectra from ChemNet embeddings:

$$\mathcal{L}_{\text{dec}} = \|\mathbf{x}_c - D(\hat{\mathbf{h}}_c)\|^2 \quad (2)$$

This decoupled architecture has profound implications for our work. Since the latent space is predetermined by ChemNet’s training on molecular structures, with no Gaussian distributional constraints, there is no guarantee that the resulting latent distribution is Gaussian. Indeed, as Figure 2 demonstrates, the ChemNet embeddings for IMS chemicals exhibit highly non-Gaussian, tendrill-like structures with non-uniform variance. This pre-defined, non-Gaussian latent geometry is the foundational motivation for our diffusion approach as we cannot assume

Gaussian sampling will work when the latent space itself is not Gaussian by construction.

The decoupled autoencoder thus provides a rich, chemically-informed 512-dimensional latent space that respects molecular structure relationships while exhibiting the complex manifold geometry that standard generation methods fail to capture.

C. Diffusion Models

Diffusion models learn to reverse a gradual noising process, drawing inspiration from non-equilibrium thermodynamics [8]. The forward process adds Gaussian noise over T timesteps:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \quad (3)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ controls the noise schedule.

Traditional diffusion models parameterize the reverse process to predict the noise ϵ added at each step [5], [8]:

$$\mathcal{L}_\epsilon = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2] \quad (4)$$

While this standard formulation excels at learning data distributions, it does not inherently preserve geometric structure when multiple classes occupy distinct regions of the manifold. Our work addresses this by augmenting noise prediction with explicit geometry-preserving objectives and by using a lower-noise schedule than typical image diffusion settings.

III. Method

A. Geometry-Preserving Diffusion in Latent Space

Our approach applies noise prediction diffusion directly to the 512-dimensional latent space learned by the ChemNet-informed autoencoder, augmented with a centroid-separation regularizer that explicitly preserves inter-class geometric structure.

1) Architecture: We employ a multi-layer perceptron with the following specifications:

- Input: Concatenation of noised latent $\mathbf{x}_t \in \mathbb{R}^{512}$, time embedding $t_{\text{emb}} \in \mathbb{R}^{512}$, SMILE embedding $\mathbf{s} \in \mathbb{R}^{512}$, and class one-hot $\mathbf{c} \in \mathbb{R}^8$
- Hidden Layers: 6 layers with 512 hidden dimensions
- Activation: SiLU (Swish) activation functions
- Output: Predicted noise $\hat{\epsilon} \in \mathbb{R}^{512}$

The time embedding uses sinusoidal positional encodings followed by a two-layer MLP, providing rich temporal context to the denoising network.

2) Training Objective: The diffusion schedule uses a linear beta schedule over $T = 50$ timesteps:

$$\beta_t = \beta_{\min} + (t - 1) \cdot \frac{\beta_{\max} - \beta_{\min}}{T - 1} \quad (5)$$

with $\beta_{\min} = 0.001$ and $\beta_{\max} = 0.02$. This low-noise regime was selected because it preserves local latent geometry while still moving samples toward a denoising-friendly diffusion space. In contrast, much larger terminal noise

(e.g., $\beta_{\max} \approx 0.2$ in many settings) can over-corrupt fine structure in this application.

Training samples uniformly over timesteps $t \sim \text{Uniform}(0, T - 1)$ and optimizes a combined objective:

1. Noise Prediction Loss: Standard DDPM objective predicting the noise added during forward diffusion:

$$\epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (6)$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon \quad (7)$$

$$\hat{\epsilon} = \epsilon_\theta(\mathbf{x}_t, t, \mathbf{s}, \mathbf{c}) \quad (8)$$

$$\mathcal{L}_{\text{noise}} = \|\epsilon - \hat{\epsilon}\|^2 \quad (9)$$

2. Geometry Regularization Loss: To preserve inter-class geometric structure, we compute predicted clean latents and enforce minimum distances between class centroids:

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\epsilon}}{\sqrt{\bar{\alpha}_t}} \quad (10)$$

$$\boldsymbol{\mu}_c = \frac{1}{|B_c|} \sum_{\mathbf{x}_i \in B_c} \hat{\mathbf{x}}_{0,i} \quad (\text{class centroids}) \quad (11)$$

$$\mathcal{L}_{\text{sep}} = \frac{1}{C(C-1)} \sum_{i \neq j} \max(0, m - \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2) \quad (12)$$

where B_c denotes samples of class c in the batch, $C = 8$ is the number of classes, and $m = 5.0$ is the margin hyperparameter.

3. Combined Objective:

$$\mathcal{L} = \lambda_{\text{noise}}\mathcal{L}_{\text{noise}} + \lambda_{\text{sep}}\mathcal{L}_{\text{sep}} \quad (13)$$

with $\lambda_{\text{noise}} = 0.8$ and $\lambda_{\text{sep}} = 0.2$, balancing generative quality with geometric preservation.

We use AdamW optimization with learning rate 5×10^{-5} , weight decay 0.01, batch size 256, and train for up to 1000 epochs with early stopping (patience=100 epochs). The learning rate follows a cosine annealing schedule.

3) Sampling via DDPM: We employ standard DDPM sampling [5], iteratively denoising from Gaussian noise. Starting from $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$, we reverse the diffusion process over all $T = 50$ timesteps:

$$\hat{\epsilon}_t = \epsilon_\theta(\mathbf{x}_t, t, \mathbf{s}, \mathbf{c}) \quad (14)$$

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\hat{\epsilon}_t}{\sqrt{\bar{\alpha}_t}} \quad (15)$$

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\epsilon}_t + \sqrt{\beta_t}\mathbf{z} \quad (16)$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ for $t > 0$ and $\mathbf{z} = 0$ at the final step.

The relatively small number of timesteps (50 vs. 1000 in typical DDPM implementations) provides efficient sampling while maintaining sufficient expressivity for the 512-dimensional latent space.

B. Class-Conditioned Multi-Chemical Generation

Rather than training separate models per chemical, we condition a single diffusion model on both molecular structure (SMILE embeddings) and chemical identity (one-hot class labels). This approach provides:

- **Parameter Efficiency:** One model handles all 8 chemicals using only 222,519 training samples
- **Shared Representations:** Common chemical features are learned once and reused across related chemicals
- **Scalability:** Adding new chemicals requires only their SMILE embeddings and class labels, without retraining from scratch

The conditioning concatenates molecular information with noised latents, allowing the model to adapt its predictions based on the target chemical’s identity and structure.

IV. Results

A. Statistical Validation

We generated 500 synthetic samples per chemical (4000 total) and evaluated statistical fidelity against 74,173 held-out test samples. Table I presents per-chemical latent space statistics, showing close matching across all chemicals.

The statistical matching (means within 0.08, standard deviations within 7.4% on average) confirms that our latent diffusion model captures first and second moments without imposing isotropic structure. Notably, JP8 exhibits high PC1 variance (84.0%), indicating a nearly one-dimensional manifold that the model reproduces.

B. Geometric Validation: Manifold Structure Preservation

Statistical matching alone is insufficient to validate generation quality: two distributions can share moments while occupying vastly different geometric structures. Figure 2 presents per-chemical PCA visualizations projecting both real and generated samples onto their first two principal components.

The visualizations reveal several critical findings:

1. **Tendril Preservation:** Generated samples (red) closely follow the elongated, non-Gaussian structures of real data (blue). Unlike Gaussian sampling, which would produce isotropic clouds, the diffusion model captures the non-uniform variance and tendril-like manifold geometry.

2. **Uniform Coverage:** Synthetic samples span almost the full range of real data, avoiding mode collapse or boundary artifacts. There are still some gaps, particularly in TEPO, in which we don’t capture the full embedding range. The model does not just interpolate between training examples but generates novel points throughout the manifold.

3. **Chemical-Specific Geometry:** Each chemical exhibits distinct manifold characteristics, JP8’s linear structure, DEB’s tight clustering, DPM’s broader distribution, all faithfully reproduced by the model.

4. **No Artificial Regularization:** Unlike VAE-generated samples, which would fill the latent space normally due to $\mathcal{N}(0, \mathbf{I})$ regularization, our generations respect the natural manifold boundaries learned by the decoupled autoencoder.

C. Role of Geometry Regularization

To validate the importance of explicit geometry regularization, we compare our approach to standard noise-prediction diffusion without the centroid-separation term (i.e., $\lambda_{\text{sep}} = 0$). Without this regularizer, the model can match first and second-order statistics while losing inter-class geometric structure: generated samples from different chemicals blend together in latent space, reducing the distinct clustering that reflects chemical similarity relationships.

The centroid-separation regularizer addresses this by explicitly encouraging class centroids to maintain minimum distances during training. This margin-based objective ($m = 5.0$) provides gradients that guide the model to respect natural class boundaries while still learning within-class variation. The key insight is that geometric structure preservation requires explicit geometric supervision; matching statistical metrics alone does not guarantee manifold faithfulness.

By combining 80% noise prediction loss with 20% geometry regularization, we achieve a balance: the noise prediction term ensures high-quality generation and distributional matching, while the regularizer preserves the geometric structure that makes different chemicals distinguishable in latent space.

D. Controlled Draws in Diffused Space

To control synthetic diversity at sampling time, we performed a draw-variance sweep in the diffused space and decoded the resulting latents to spectra. Specifically, we generated synthetic sets using Gaussian draw scales $\sigma \in \{1.0, 1.5, 2.0\}$, corresponding to the datasets later evaluated as std1.0, std1.5, and std2.0.

Figure 3 shows how this draw scale changes the generated support relative to real data. Lower draw scale ($\sigma = 1.0$) yields conservative samples concentrated near dense manifold regions, while higher scales ($\sigma = 1.5, 2.0$) increase spread and introduce broader coverage. This controlled stochasticity provides a practical knob for balancing fidelity and diversity without retraining the model.

E. Low-Data Classifier Utility (Supporting Evidence)

Although this paper is method-centered, we include a downstream check to verify that preserving latent geometry leads to useful synthetic data in low-data training regimes. Using untuned Random Forest and MLP classifiers with mixed real/synthetic training sets, we compute best-minus-real improvements across 50 points-per-class budgets.

The stronger gains for MLP suggest that preserving non-Gaussian latent structure does not only improve geometric fidelity; it can also improve practical learning outcomes when labeled real data is scarce.

TABLE I
Geometry-Preserving Latent Diffusion: Per-Chemical Latent Statistics

Chemical	Mean		Standard Deviation		PCA Variance	
	Real	Generated	Real	Generated	PC1	PC2
DEB	0.02	0.02	10.67	10.41	51.4%	21.4%
DEM	-0.03	-0.05	10.07	9.96	48.6%	28.0%
DMMP	-0.22	-0.17	17.73	16.60	41.5%	24.7%
DPM	-0.01	0.10	23.75	20.73	37.5%	23.7%
DtBP	-0.38	-0.31	18.48	16.89	55.8%	15.8%
JP8	-0.06	-0.08	11.42	11.21	84.0%	7.0%
MES	0.19	0.21	14.62	13.98	49.9%	23.3%
TEPO	-0.47	-0.32	21.84	18.04	45.4%	20.3%
Mean Error	0.08		1.53 (7.4% relative)		—	

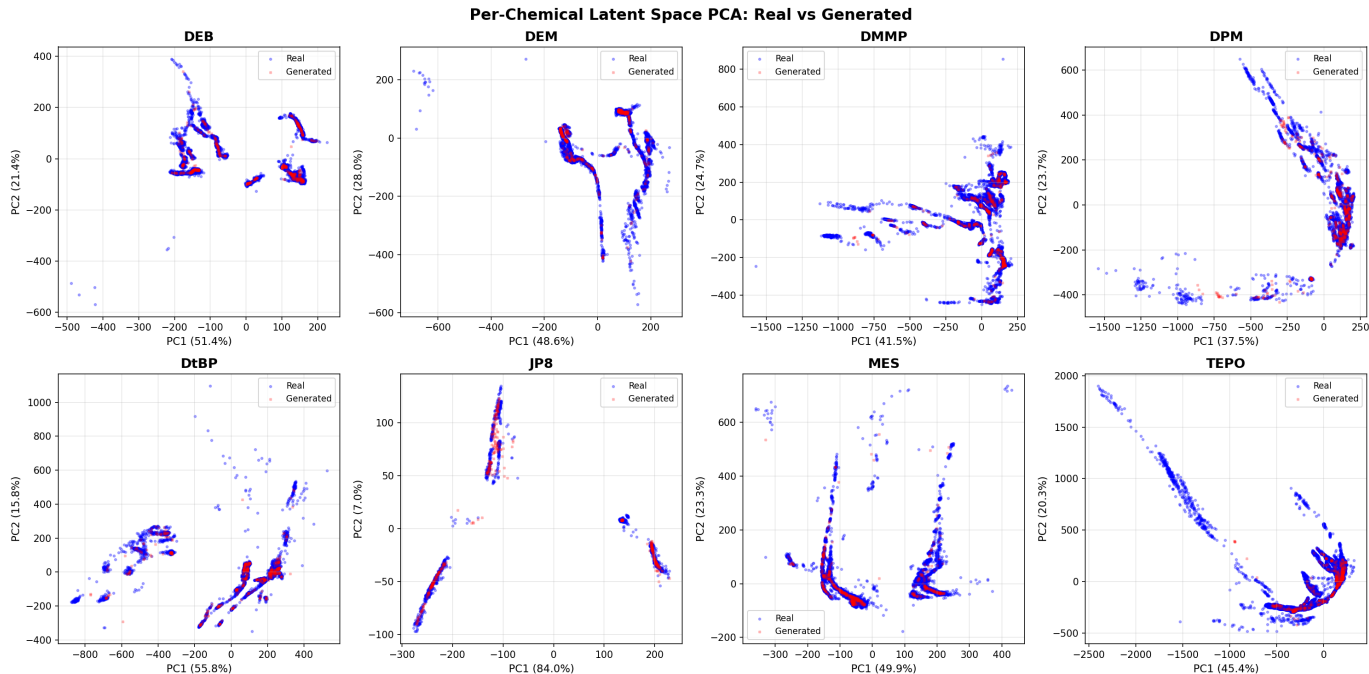


Fig. 2. Per-chemical PCA visualization of real (blue) vs. generated (red) latent samples. Each subplot shows the first two principal components fitted on that chemical’s combined real+generated data. The proposed latent diffusion model preserves the tendril-like manifold geometry characteristic of IMS latent representations. Explained variance ratios indicate chemical-specific dimensionality: JP8’s near-linear structure (84.0% PC1) contrasts with DPM’s more distributed representation (37.5% PC1, 23.7% PC2). Generated samples follow these natural geometric structures rather than collapsing to isotropic Gaussian distributions, demonstrating that geometry-aware training maintains inter-class structure while respecting manifold geometry.

TABLE II
Downstream low-data classification lift (best-minus-real accuracy).

Model	Std	Mean Δ	Improved
RF	1.0	+0.0097	20/50
RF	1.5	+0.0061	30/50
RF	2.0	+0.0080	28/50
MLP	1.0	+0.0129	25/50
MLP	1.5	+0.1519	49/50
MLP	2.0	+0.1304	50/50

dimensional spectra exhibit:

- Chemical-Specific Peak Patterns: Generated spectra show characteristic peak structures matching the target chemical class
- Realistic Noise Characteristics: Baseline fluctuations and intensity distributions match real measurements
- Physical Validity: No negative intensities or physically implausible artifacts

F. Decoded Spectra Quality

After generating latent samples, we decode them to full IMS spectra using the trained decoder. The resulting 1676-

These properties confirm that preserving manifold structure in latent space translates to high-quality, scientifically plausible spectra in the original measurement space.

Per-Chemical Latent Space PCA: Real vs Generated with Varying Initial Noise
Real (red circles) | std=1.0 (blue squares) | std=1.5 (green triangles) | std=2.0 (orange diamonds)

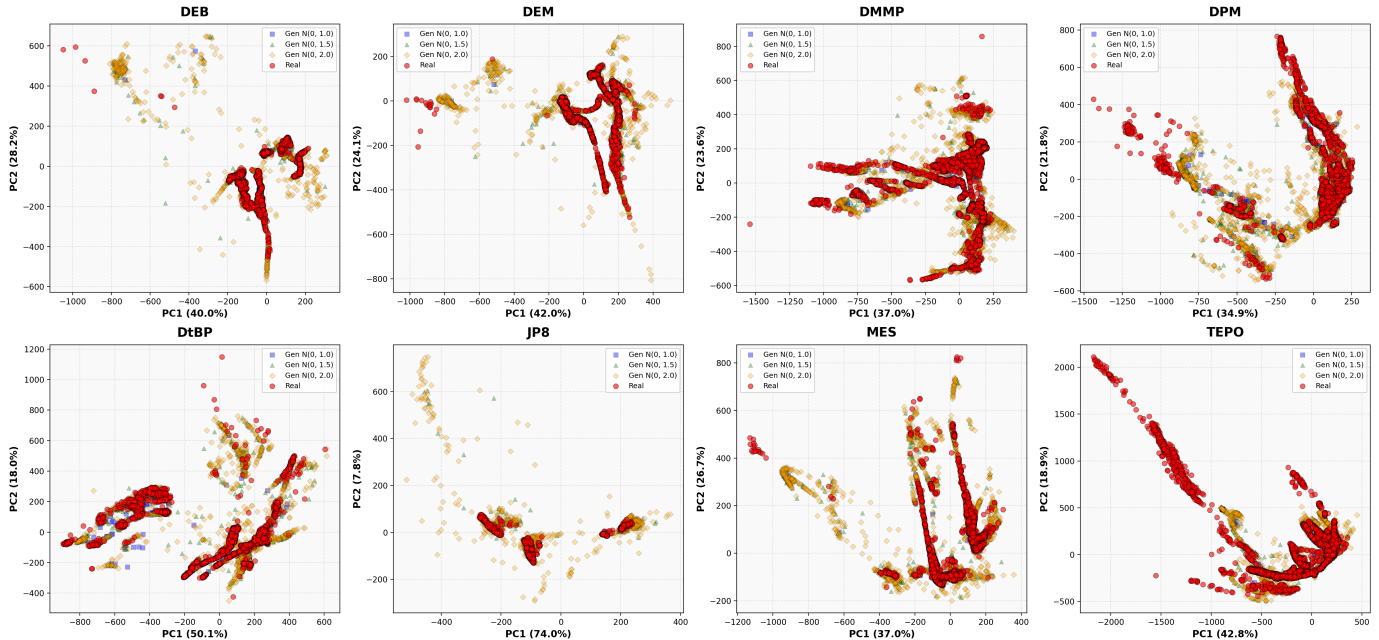


Fig. 3. Effect of draw standard deviation in diffused space on per-chemical latent coverage. Increasing draw scale from $\sigma = 1.0$ to $\sigma = 2.0$ expands generated support relative to real-data structure, providing a direct diversity-control mechanism at sampling time.

V. Discussion

A. The Importance of Natural Manifold Structure

Our results demonstrate that respecting the natural geometry of learned representations is essential for high-quality synthetic data generation. The comparison between our geometry-preserving latent diffusion setup and standard diffusion without geometric constraints reveals that how statistical moments are matched matters as much as that they are matched.

VAE-style Gaussian regularization represents the opposite extreme: by forcing $q(\mathbf{z}|\mathbf{x}) \approx \mathcal{N}(0, \mathbf{I})$ through KL divergence penalties, we gain sampling convenience at the cost of destroying natural manifold structure. Our approach maintains this structure by augmenting standard diffusion with explicit geometric objectives.

B. Geometry Regularization During Training

The superior performance of this approach stems from explicit geometric supervision during training. Standard noise prediction models learn to reverse the diffusion process:

$$\epsilon_\theta : \mathbb{R}^d \times [0, T] \times \mathcal{C} \rightarrow \mathbb{R}^d \quad (17)$$

mapping noisy latents, timesteps, and conditioning $\mathcal{C}$ to noise predictions. Without geometric constraints, this objective can produce samples that match statistical moments while collapsing inter-class structure.

Our separation loss provides complementary supervision by operating on the predicted clean latents:

$$\mathcal{L}_{\text{sep}} : \{\hat{\mathbf{x}}_0^{(i)}, c_i\}_{i=1}^B \rightarrow \mathbb{R}^+ \quad (18)$$

penalizing insufficient distance between class centroids. This margin-based constraint ($m = 5.0$) encourages the model to maintain distinct chemical clusters while learning distributional structure. The combined objective (80% noise + 20% separation) balances generative quality with geometric fidelity.

C. Implications for Low-Data Scientific Domains

In domains like IMS, where data acquisition is expensive, our findings suggest that:

1. Standard Autoencoders Suffice: VAE-style Gaussian regularization is unnecessary, standard autoencoders learn rich, geometrically meaningful representations without distributional constraints.
2. Geometric Supervision Matters: Augmenting standard diffusion with explicit geometric objectives (centroid-separation regularization) preserves manifold structure that statistical objectives alone cannot guarantee.
3. Multi-Task Learning Works: A single class-conditioned diffusion model can generate for multiple related chemicals, leveraging shared structure while maintaining chemical-specific features through separation guidance.

D. Limitations and Future Work

While our approach achieves strong results on IMS data, several directions warrant investigation:

Uncertainty Quantification: Developing principled methods to estimate confidence for generated samples would enable selective augmentation strategies, filtering low-confidence synthetics before downstream use.

Decoded IMS Spectra: Real vs Diffusion vs Gaussian
(Lines show mean, shaded show individual samples)

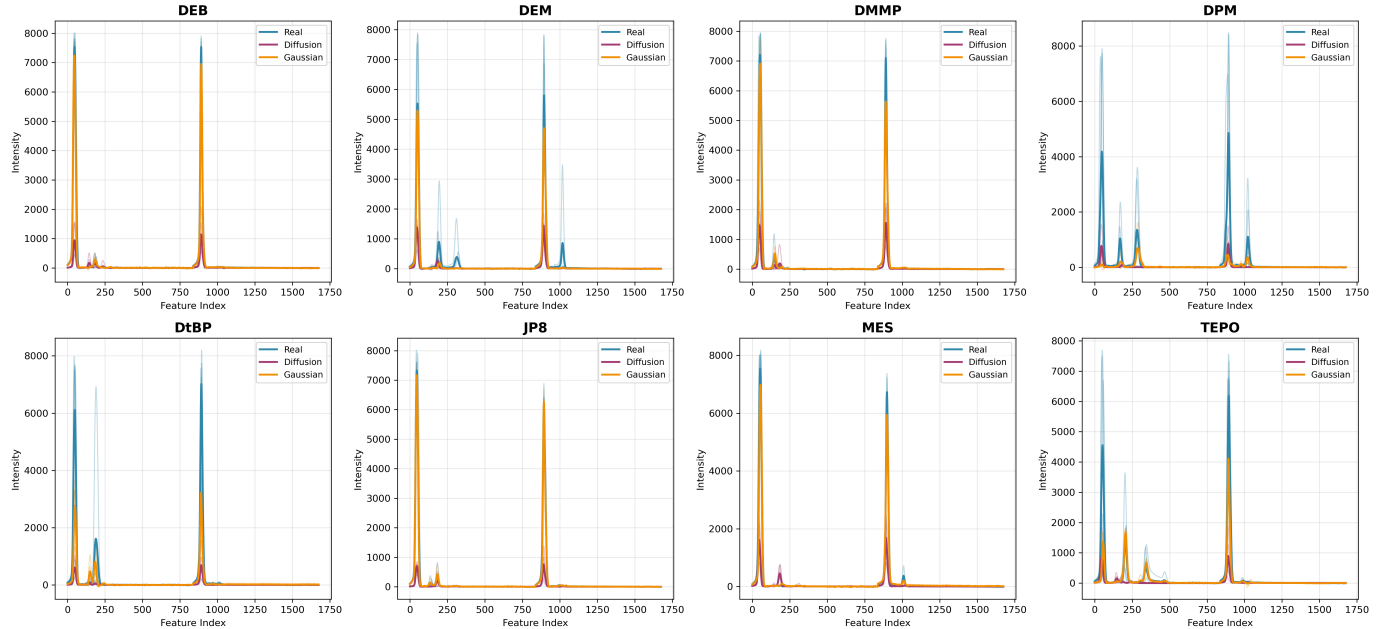


Fig. 4. Real vs. generated spectra across all eight chemicals after decoding from latent diffusion samples. Generated spectra recover class-specific peak structure and realistic intensity profiles, indicating that latent-space geometry preservation transfers to the measurement space.

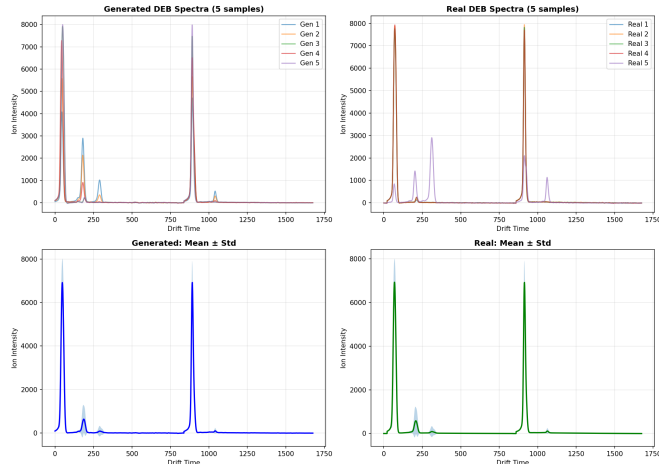


Fig. 5. DEB-only real vs. generated spectra comparison. The generated DEB signals follow the same dominant peak locations and local profile shape as real measurements.

Conditional Generation Beyond Classes: Extending conditioning to continuous chemical properties (molecular weight, functional groups, volatility) could enable interpolation between known chemicals and targeted generation of spectra for novel compounds.

Ablation Studies: Systematic investigation of the loss weighting hyperparameters (λ_{noise} , λ_{sep}) and margin value (m) would provide insights into the trade-off between generative quality and geometric preservation across different datasets.

Theoretical Analysis: Formalizing the conditions under which geometry-regularized latent diffusion preserves geometric structure would provide theoretical grounding for the empirical success of margin-based geometric regularization.

VI. Conclusion

This work demonstrates that latent diffusion in a pre-given embedding space can preserve the natural manifold structure of chemical spectral data while matching target distributions. Applied to ion mobility spectrometry data across 8 chemicals and 296,692 samples, our approach achieves strong statistical matching while maintaining complex geometric features that naive Gaussian sampling destroys.

The key insight is that learned representations naturally occupy non-Gaussian manifolds reflecting intrinsic data structure. Rather than constraining these representations through VAE-style regularization, we show that augmenting standard diffusion with explicit geometric supervision, via a margin-based centroid-separation regularizer, preserves inter-class structure while enabling within-class variation. By combining 80% noise prediction with 20% geometry regularization, we achieve a principled balance between generative quality and geometric fidelity.

For low-data scientific domains where preserving physical structure is as important as matching statistical moments, our findings suggest that diffusion in pre-given latent space can preserve useful information while matching target distributions.

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